

Algebra II with Trigonometry

Chapter 5.3

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4401269_alg-2trig-h-chp-5-3-note-docx.pdf](#)

1. **Reference Angle:** What is the reference angle for 210° ?
2. **Quadrant Sign:** In which quadrant is θ if $\sin \theta < 0$ and $\cos \theta > 0$?
3. **Exact Value:** What is $\sin 150^\circ$?
4. **Coterminal Thinking:** What is $\cos 390^\circ$?
5. **Negative Angle:** What is $\tan(-45^\circ)$?
6. **Undefined Function:** At 90° , which is undefined: $\sin \theta$, $\cos \theta$, or $\tan \theta$?
7. **Coordinate Definition:** If a point on the terminal side of θ is $(3, 4)$, what is $\sin \theta$?
8. **Reciprocal Identity:** If $\cos \theta = \frac{1}{2}$, what is $\sec \theta$?
9. **Pythagorean Identity:** If $\sin \theta = \frac{3}{5}$ and θ is acute, what is $\cos \theta$?
10. **Triangle Area:** A triangle has sides 6 and 8 with included angle 30° . What is its area?